Modulating inflammatory pathways in CACD organotypic retinal explant cultures using anti-inflammatory mediator and microRNA-based targeting strategies

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Central areolar choroidal dystrophy (CACD) is an inherited retinal disease characterized by progressive photoreceptor degeneration with no effective treatment. Increasing evidence implicates early dysregulation of inflammatory and complement pathways in disease progression.

Transcriptomic profiling (mRNA sequencing and microRNA) was performed across disease progression at 1, 3, 6, and 9 months in a CRISPR-generated Prph2 knock-in mouse model carrying the PRPH2 p.Arg195Leu mutation. This analysis identified early alterations in inflammatory and complement-related pathways, with a marked upregulation observed between 1 and 3 months. Based on these findings, ex vivo retinal explant cultures from this model were used to test therapeutic modulation. Explants were treated with the pro-resolving lipid mediator Lipoxin B4 (LBX4). Complement pathway activity was assessed by gene expression analysis (qPCR), with specific quantification of C3 and C4b. Treatment toxicity was evaluated by flow cytometry through the assessment of apoptosis and cell death.

Transcriptomic analysis revealed early activation of complement and inflammatory pathways prior to overt degeneration, peaking between 1 and 3 months. LBX4 treatment induced a reduction in complement related markers, with decreased expression of C3 and C4b observed at 100 nM and 1000 nM. However, responses in retinal explants after 24 h differed from those observed in vivo at baseline (T0), suggesting a partial loss of complement regulation under ex vivo conditions. These observations suggest that complement modulation may be reduced in retinal explants after 24 h, emphasizing the importance of optimizing the experimental context. Ongoing experiments are therefore focusing on earlier disease stages and alternative experimental models to better capture the initial complement dysregulation. Flow cytometry analyses indicated no major increase in apoptosis or cell death under these conditions. In parallel, a microRNA-based approach is being developed to further modulate inflammatory responses.

These results identify early complement dysregulation as a targetable process in CACD and demonstrate the potential of Lipoxin B4 as a modulator of inflammatory pathways. The reduced effect observed after 24 h in retinal explants highlights critical limitations related to disease stage and experimental context, guiding ongoing optimization and validation in early-stage and alternative models. Flow cytometry analyses further indicate no significant increase in apoptosis or cell death in retinal explants. This work reinforces the translational relevance of organotypic retinal systems for therapeutic development.